Algebraic stabilizations on anisotropic meshes: accuracy, nonlinear solvers and the DMP mesh conditions

galerkit

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Abstract

For convection–diffusion problems whose layers are known in advance, layer-adapted meshes such as Shishkin meshes use cells that are small across the layer and long along it. We study the eight algebraic stabilizations of galerkit (AFC with the BJK, Kuzmin and monotone convex limiters, MUAS, SMUAS, BBK, F-BBK, S-BBK) on uniform and Shishkin meshes with aspect ratios up to $6 \cdot 10^7$, for three problems with exponential and parabolic layers and $\varepsilon = 10^{-4}, 10^{-8}$: 638 runs. Shishkin meshes resolve the layers and reduce the L^2 error by up to four orders of magnitude for all methods. On these meshes, however, the Galerkin method is already stable, and the algebraic stabilization costs accuracy: for a problem with layers in both directions its errors are 5–40 times larger than those of Galerkin and SUPG, with first-order convergence away from the layers. The nonlinear problems become harder: the iteration numbers of the fixed point rhs grow with the anisotropy, and 62 of the 510 runs did not converge with any of three solvers. The DMP mesh conditions of the Kuzmin, MC and F-/S-BBK methods hold on Shishkin meshes split into two right triangles per cell and fail on the non-Delaunay crossed splitting; no DMP violation was observed even there.

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1 Questions

If the position of the layers is known, it makes sense to use mesh cells that are small in the direction of the convection (across an exponential layer) and large orthogonal to it; Shishkin meshes are the standard example. For the algebraic stabilizations we ask:

1. **Accuracy** of the solutions on such meshes, compared with uniform meshes and with Galerkin and SUPG.

2. **Efficiency**: the number of iterations needed to solve the nonlinear problem.

3. **DMP**: some methods satisfy the discrete maximum principle on all simplicial meshes, others only under a mesh condition. Are these conditions satisfied on anisotropic meshes, and is the DMP violated where they are not?

2 Set-up

Methods. P_1 elements, $\sum_j (a_{ij} + b_{ij}(u))u_j = g_i$ with the Galerkin matrix A and a symmetric stabilization matrix $B(u)$; the reaction term is not lumped.

method	galerkit	sufficient DMP condition (`dmp_certificate`)
AFC, BJK limiter	`afc`	none (any simplicial mesh)
MUAS, SMUAS	`muas`, `smuas`	none
BBK	`bbk`	none
AFC, Kuzmin limiter	`afc_kuzmin`	$\min\{a_{ij}, a_{ji}\} \le 0$ on every edge with a free node
AFC, monotone convex limiter	`afc_mc`	Delaunay type (Kuzmin condition checked as a proxy)
F-BBK, S-BBK	`fbbk`, `sbbk`	$\max\{a_{ij}, a_{ji}\} \le \tau_E = C_0\lvert E\rvert$

SUPG and the Galerkin method serve as references.

Problems (layers known), on $\Omega = (0,1)^2$:

- `exp_layer`: $\boldsymbol{b} = (1,0)$, $c = f = 0$, $u = (1 - e^{(x-1)/\varepsilon})/(1 - e^{-1/\varepsilon}) \in [0,1]$, an exponential layer at $x = 1$; with $f = c = 0$ the local DMP applies, so its violations can be counted node by node. $\varepsilon = 10^{-4}, 10^{-8}$.

- `boundary_layers`: $\boldsymbol{b} = (2,3)$, $c = 1$, exact solution with exponential layers at $x = 1$ and $y = 1$; $\varepsilon = 10^{-8}$.

- `parabolic_layers`: $\boldsymbol{b} = (1,0)$, $f = 1$, $u_b = 0$: exponential layer at $x = 1$, parabolic layers at $y = 0, 1$; no exact solution, but $0 \le u \le x \le 1$. $\varepsilon = 10^{-8}$.

Meshes. $n \times n$ cells, $n = 16, 32, 64, 128$: uniform ("/" diagonals) and Shishkin meshes with $n/2$ cells in each layer region of width $\tau = 2\varepsilon/\beta \ln n$ for an exponential layer (β the convection component across it) and $n/4$ cells in each region of width $\tau = 2\sqrt{\varepsilon}\ln n$ for the parabolic layers. Each Shishkin cell is split by "/", "\" or both diagonals (Figure 1). With two triangles per cell all triangles are right-angled and the mesh is Delaunay; the crossed splitting has angles up to 180° in the layer cells and is not Delaunay (Table 1).

Solvers and measures. For every algebraic method: the fixed point rhs (`fp_rhs`), the standard solver for these schemes, with at most 3000 iterations (efficiency); if it does not converge (Euclidean residual $> 10^{-10}$), galerkit's Newton portfolio, and for $n \le 32$ in addition the Anderson-accelerated fixed point with up to 20 000 iterations. Accuracy: the L^2 error, the maximal nodal error, and the maximal nodal error away from the layer regions. The energy norm is not used: on uniform meshes the quadrature does not see layers of width ε. DMP: the mesh condition of the method checked edge by edge, the local DMP for `exp_layer`, and the bounds of the exact solution.

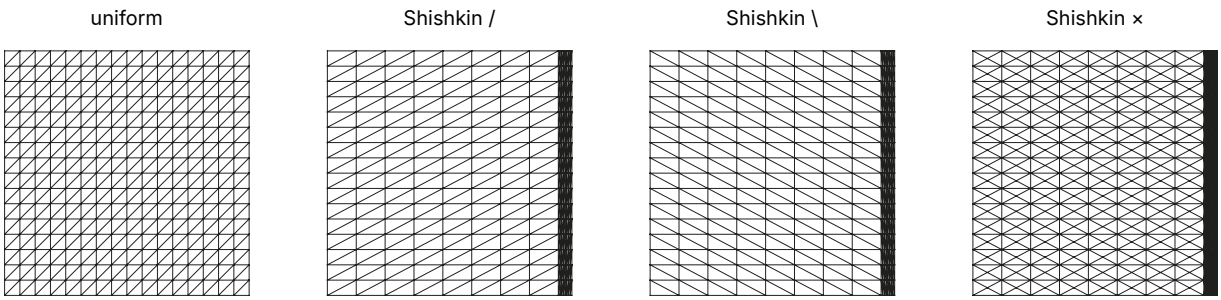


Figure 1: The mesh types for `parabolic_layers`, $n = 16$; drawn for $\varepsilon = 10^{-2}$ so that the layer cells are visible (at $\varepsilon = 10^{-8}$ the layer regions have widths of order 10^{-7} and 10^{-3}).

Table 1: The meshes at $n = 128$: nodes, largest aspect ratio (longest edge over its height), shortest edge, largest angle, and the edges with a positive off-diagonal entry of the P_1 Laplace matrix (not Delaunay).

case	mesh	nodes	aspect	$h_{\min}$	max angle	Laplace > 0
`exp_layer`, $\varepsilon = 10^{-4}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin /	16641	5.2e2	1.5e−5	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin \	16641	5.2e2	1.5e−5	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin ×	33025	1.0e3	1.5e−5	179.8	16320
`exp_layer`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin /	16641	5.2e6	1.5e−9	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin \	16641	5.2e6	1.5e−9	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	1.0e7	1.5e−9	180.0	16320
`boundary_layers`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin /	16641	3.1e7	5.1e−10	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin \	16641	3.1e7	5.1e−10	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	6.2e7	4.6e−10	180.0	12224
`parabolic_layers`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin /	16641	1.0e7	1.5e−9	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin \	16641	1.0e7	1.5e−9	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	2.1e7	1.5e−9	180.0	16320

3 Accuracy

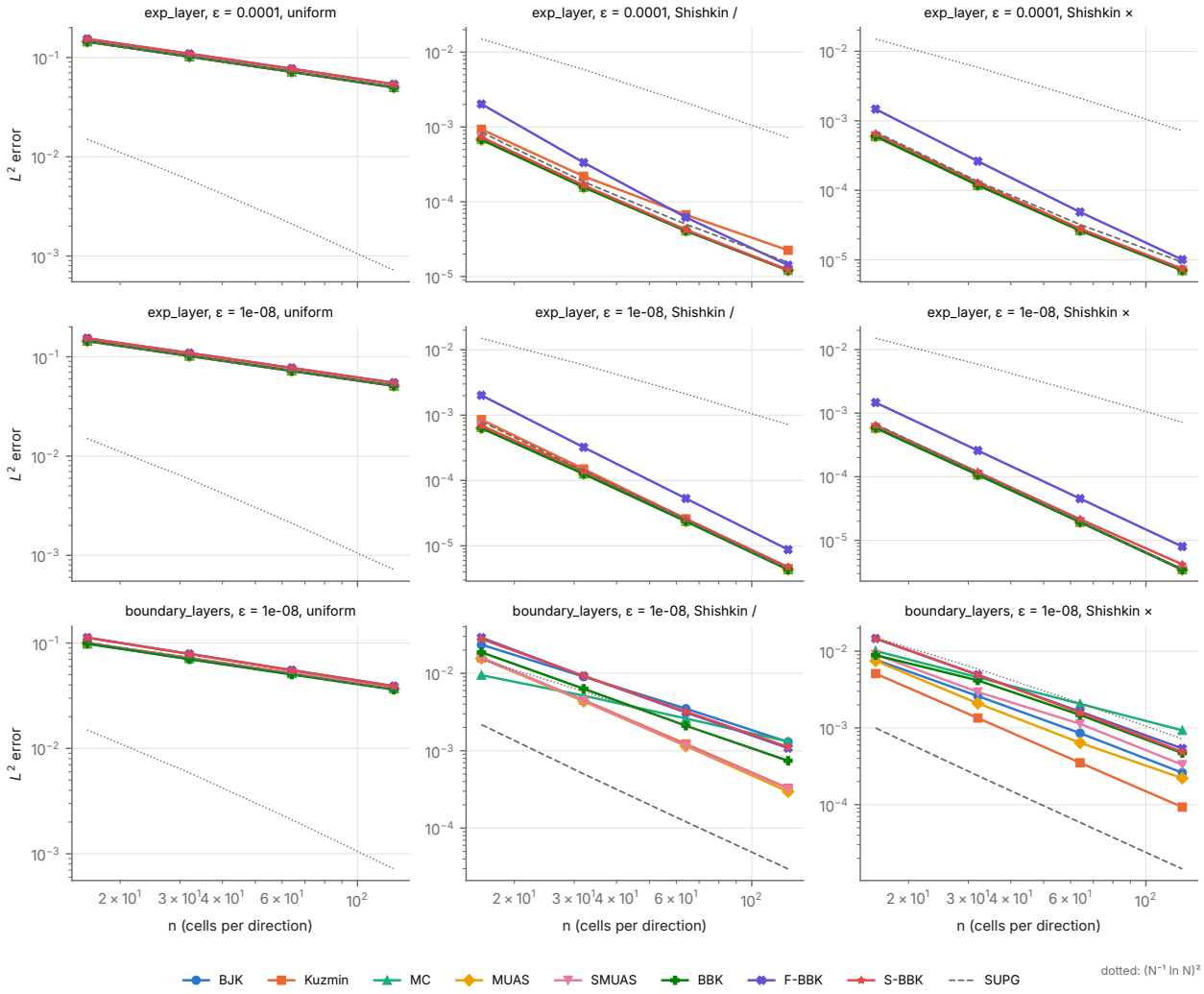


Figure 2: L^2 error against n on uniform, Shishkin "/" and crossed Shishkin meshes.

Table 2: `exp_layer`, $\varepsilon = 10^{-4}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	1.6e0	0.51	1.6e−1	7.8e−4	1.61	2.3e−5	9.4e−4	1.53	8.1e−6
SUPG	2.3e−14	-1.86	5.0e−2	1.6e−13	-2.56	1.5e−5	8.7e−4	1.45	9.3e−6
BJK	1.6e−14	-1.04	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
Kuzmin	1.5e−14	-1.01	5.0e−2	3.0e−3	0.95	2.2e−5	9.4e−4	1.53	7.1e−6
MC	1.5e−14	-1.01	5.0e−2	7.8e−4	1.57	1.2e−5	9.4e−4	1.53	7.1e−6
MUAS	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
SMUAS	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
BBK	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
F-BBK	1.2e−1	-0.01	5.4e−2	7.9e−4	1.51	1.4e−5	9.4e−4	1.53	1.0e−5
S-BBK	1.2e−1	-0.01	5.4e−2	7.8e−4	1.55	1.2e−5	9.4e−4	1.53	7.4e−6

Table 3: `exp_layer`, $\varepsilon = 10^{-8}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	2.4e3	1.44	6.1e2	8.4e−4	1.59	7.2e−5	7.2e−3	1.15	4.7e−4
SUPG	2.8e−14	-1.32	5.1e−2	7.2e−14	-1.65	4.4e−6	8.7e−4	1.45	3.5e−6
BJK	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
Kuzmin	1.5e−14	-1.40	5.1e−2	3.0e−3	0.95	4.5e−6	9.4e−4	1.53	3.5e−6
MC	1.5e−14	-1.40	5.1e−2	7.8e−4	1.57	4.3e−6	9.4e−4	1.53	3.5e−6
MUAS	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
SMUAS	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
BBK	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
F-BBK	1.2e−1	-0.00	5.5e−2	7.9e−4	1.51	8.7e−6	9.4e−4	1.53	8.0e−6
S-BBK	1.2e−1	-0.00	5.5e−2	7.8e−4	1.55	4.7e−6	9.4e−4	1.53	4.2e−6

Table 4: `boundary_layers`, $\varepsilon = 10^{-8}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	7.0e0	0.06	1.6e0	1.5e−2	0.78	6.5e−5	1.9e−3	1.34	3.3e−5
SUPG	2.9e−1	-0.04	3.6e−2	1.7e−2	0.89	3.0e−5	3.2e−3	1.30	1.5e−5
BJK	5.3e−2	0.82	3.6e−2	2.1e−2	0.96	1.3e−3	1.3e−2	0.90	2.6e−4
Kuzmin	1.6e−2	0.98	3.7e−2	4.5e−2	0.64	3.3e−4	3.1e−2	0.61	9.3e−5
MC	2.5e−2	0.94	3.7e−2	4.0e−2	0.59	1.3e−3	5.9e−2	0.56	9.3e−4
MUAS	1.6e−2	0.98	3.7e−2	2.4e−2	0.72	3.0e−4	6.5e−2	0.57	2.2e−4
SMUAS	1.6e−2	0.97	3.7e−2	3.2e−2	0.19	3.2e−4	2.9e−2	1.43	3.3e−4
BBK	6.8e−2	0.74	3.6e−2	3.0e−2	0.67	7.4e−4	3.2e−2	0.56	4.7e−4
F-BBK	1.7e−1	0.21	3.9e−2	2.9e−2	0.97	1.1e−3	1.8e−2	0.97	5.4e−4
S-BBK	1.7e−1	0.21	3.9e−2	3.7e−2	0.69	1.1e−3	2.9e−2	0.69	4.9e−4

- **Shishkin meshes resolve the layers** (Figure 2). For `exp_layer` the L^2 error at $n = 128$ drops from $5 \cdot 10^{-2}$ on the uniform mesh to about $4 \cdot 10^{-6}$ ($\varepsilon = 10^{-8}$) for every method, with nearly the rate $(N^{-1} \ln N)^2$ and independently of ε. On the uniform mesh the nodal values of this layer, aligned with the mesh, are exact for most methods although the layer is not resolved; the maximal nodal error alone is therefore not a measure of quality here.
- **The stabilization costs accuracy where it is not needed.** For `boundary_layers` on the "/" Shishkin mesh the Galerkin method (L^2 error $6.5 \cdot 10^{-5}$) and SUPG ($3 \cdot 10^{-5}$) converge with order 2. The algebraic methods are 5–40 times less accurate: Kuzmin and MUAS $3 \cdot 10^{-4}$ (order about 1.9; SMUAS the same, but from unconverged iterates), BJK and BBK 0.7–$1.3 \cdot 10^{-3}$ (order 1.5), MC $1.3 \cdot 10^{-3}$ (order 1). Away from the layers their maximal error, about 10^{-2}, converges only with order 0.7–1, against 1.6–1.8 for Galerkin and SUPG. On a mesh that resolves the layers, the Galerkin method is stable; the artificial diffusion of the limiters then only adds a first-order consistency error.
- **Splitting.** "/" and "\" give nearly the same errors. The crossed splitting doubles the number of nodes and gives the same accuracy for `exp_layer`; for `boundary_layers` it changes the ordering of the methods (Kuzmin, MUAS best).
- F-BBK is the least accurate method for `exp_layer` (about twice the error of the others).

4 Efficiency of the nonlinear solvers

Table 5: Iterations of the fixed point rhs at $n = 128$ (u: uniform, /: Shishkin /, ×: Shishkin ×); in colour: not converged within the limit (residual $> 10^{-10}$).

	`exp_layer`, $\varepsilon = 10^{-4}$			`exp_layer`, $\varepsilon = 10^{-8}$			`boundary_layers`, $\varepsilon = 10^{-8}$			`parabolic_layers`, $\varepsilon = 10^{-8}$		
method	u	/	×	u	/	×	u	/	×	u	/	×
BJK	0	997	3000	0	1075	3000	3000	3000	3000	3000	3000	3000
Kuzmin	0	95	355	0	95	355	176	964	3000	265	358	3000
MC	0	454	3000	0	461	3000	3000	3000	3000	3000	3000	3000
MUAS	0	449	3000	0	456	3000	176	3000	3000	291	1298	3000
SMUAS	0	406	3000	0	408	3000	176	3000	3000	252	3000	3000
BBK	0	1426	3000	0	1465	3000	3000	3000	3000	3000	3000	3000
F-BBK	246	3000	3000	251	3000	3000	3000	3000	3000	3000	3000	–
S-BBK	246	3000	3000	251	3000	3000	3000	3000	3000	3000	3000	–

- **The iteration numbers grow with the anisotropy.** Kuzmin, `exp_layer`, $\varepsilon = 10^{-8}$, $n = 128$: 0 iterations on the uniform mesh (the low-order start is already the solution), 95 on the Shishkin "/" mesh, 355 on the crossed one. MUAS, SMUAS and MC need 400–460 iterations on the "/" Shishkin mesh and do not converge within 3000 on the crossed one.
- **Robustness of fp_rhs** (converged runs out of 64 per method): Kuzmin 62, MUAS 56, MC 52, BBK 36, BJK 29, F-BBK 19, S-BBK 17. SMUAS converges on every uniform mesh but in only 22 of the 48 Shishkin runs.
- **Not converged at all.** With the Newton portfolio and the Anderson iteration as fallbacks, 62 of the 510 runs still end with a residual above 10^{-8}: BJK 25, F-BBK 13, S-BBK 12 (mostly `parabolic_layers`) and SMUAS 12 (all three Shishkin meshes of `boundary_layers`). The Anderson iteration converged in none of the 28 cases it was tried on. The errors reported for these runs are those of the best iterate.

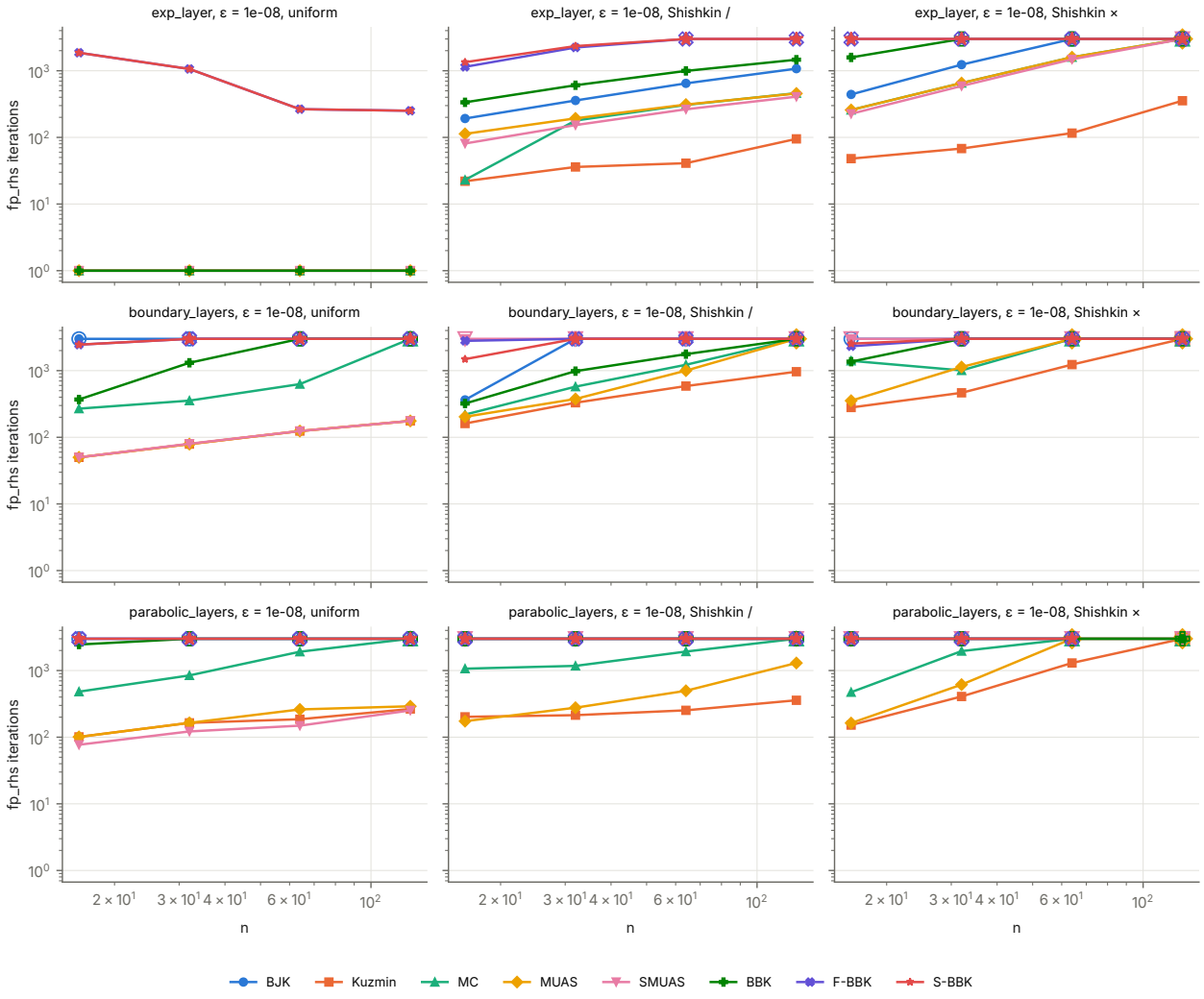


Figure 3: Iterations of the fixed point rhs; open circles: not converged within 3000 iterations.

5 The DMP and the mesh conditions

Table 6: DMP at $n = 128$: "ok" if the sufficient mesh condition of the method holds on every edge, otherwise the number of edges where it fails (colour); for `exp_layer` ($f = c = 0$) a second number gives the nodes violating the local DMP; †: the solution leaves the bounds of the exact solution by more than 10^{-8}.

	`exp_layer`, $\varepsilon = 10^{-4}$			`exp_layer`, $\varepsilon = 10^{-8}$			`boundary_layers`, $\varepsilon = 10^{-8}$			`parabolic_layers`, $\varepsilon = 10^{-8}$		
method	u	/	×	u	/	×	u	/	×	u	/	×
BJK	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok†	ok†
Kuzmin	ok	ok	8192	ok	ok	8192	ok	ok	12224	ok	ok†	8192
MC	ok	ok	8192	ok	ok	8192	ok	ok	12224	ok	ok	8192
MUAS	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok
SMUAS	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok
BBK	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok
F-BBK	ok	ok	8064	ok	ok	8064	ok	ok	12224	ok	ok†	–
S-BBK	ok	ok	8064	ok	ok	8064	ok	ok	12224	ok	ok†	–

- **Two right triangles per cell: the conditions hold.** On the uniform meshes and on the "/" and "\" Shishkin meshes the Kuzmin, MC and F-/S-BBK conditions hold on every edge, for all problems. The diffusion matrix of right triangles has no positive off-diagonal entries, and the non-lumped reaction term of `boundary_layers` ($c = 1$) does not break $\min\{a_{ij}, a_{ji}\} \le 0$, since convection dominates.
- **Crossed splitting: the conditions fail.** Where the layer cells are split into four triangles, the obtuse angles (up to 180°) make diffusion entries positive: the Kuzmin/MC condition fails on 128 to 12 224 edges, the F-/S-BBK condition at the larger n (8 064 edges for `exp_layer` at $n = 128$).
- **No observed DMP violation.** For `exp_layer`, where the local DMP applies, no method violates it on any mesh, including Kuzmin and MC on the crossed meshes where their condition fails. For `parabolic_layers` no method undershoots $u \ge 0$.
- **Overshoots of the exact bound.** On the "/" and "\" Shishkin meshes of `parabolic_layers`, BJK, Kuzmin, MUAS, BBK, F-BBK and S-BBK exceed the bound $u \le 1$ of the exact solution by up to 2–9 %; this is not a DMP violation ($f = 1$), but non-physical. MC and SMUAS stay below $1 + 6 \cdot 10^{-4}$, and no method overshoots on the crossed meshes.

6 Conclusions and next steps

1. On layer-adapted meshes all algebraic stabilizations resolve the layers, but they are not needed for stability there, and they reduce the accuracy (first order away from the layers). A natural remedy is to switch the stabilization off where the mesh resolves the layer, e.g. by limiting only on edges with mesh Péclet number $|\boldsymbol{b}|h_E/\varepsilon > 1$.
2. The nonlinear problems become harder on anisotropic meshes. Kuzmin and MUAS remain the most robust; SMUAS, which is the most accurate method on general isotropic meshes, often fails to converge on Shishkin meshes.
3. The DMP conditions of the restricted methods are satisfied on Shishkin meshes split into right triangles and violated on crossed splittings, without observed DMP violations; the guaranteed

methods need no condition. For anisotropic meshes the splitting into right triangles is therefore the safe choice.

4. Open: Bakhvalov-type meshes, anisotropic meshes not aligned with the axes, 3D, the solver failures of SMUAS on `boundary_layers`, and the role of σ in the transition point.

Reproducibility. galerkit branch `feature/anisotropic-afc`, folder

`examples/convection_diffusion/2d/anisotropic_afc`:

`run_aniso.py` (the runs), `make_report.py` (these tables and figures) and `results.json` (the 638 runs). F-BBK and S-BBK on the crossed Shishkin mesh of `parabolic_layers` at $n = 128$ were stopped after more than an hour and are missing.